

HW05

第05章 连接及连接件

参考答案

所有作业要求手写

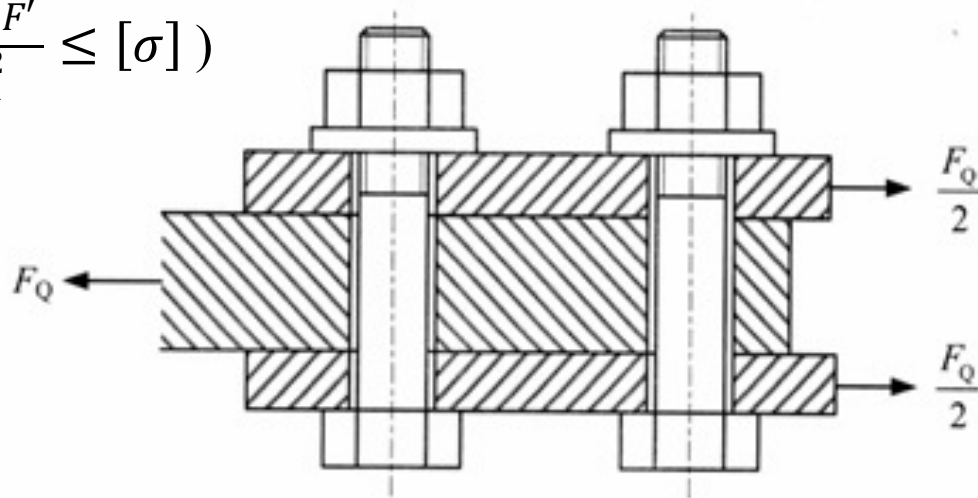
Autumn 2024

HW 05.1

- 图示螺栓连接中采用 2 个 M20 ($d_1 = 17.3\text{mm}$) 的螺栓，其许用拉应力为 $[\sigma]=160\text{MPa}$ ，被连接件接合面间的摩擦系数 $\mu = 0.2$ ，若考虑摩擦传力的可靠系数 $K_f = 1.2$ ，试计算该连接允许传递的静载荷 R 。

(普通螺栓拉伸强度条件 $\sigma_{ca} = \frac{4 \times 1.3 F'}{\pi d_1^2} \leq [\sigma]$)

- The bolted connection shown in the figure is made with 2 bolts of M20 ($d_1 = 17.3\text{mm}$) with an allowable tensile stress of $[\sigma]=160\text{MPa}$, and the coefficient of friction between the joint surfaces of the connected parts $\mu=0.2$. If the reliability coefficient of friction transmission $K_f = 1.2$ is taken into account, try to calculate the permissible static load to be transferred by this connection. R . (Common bolt tensile strength condition $\sigma_{ca} = \frac{4 \times 1.3 F'}{\pi d_1^2} \leq [\sigma]$)



HW 05.1

解：由普通螺栓拉伸强度条件 $\sigma_{ca} = \frac{4 \times 1.3 F'}{\pi d_1^2} \leq [\sigma]$ 可得螺栓允许承受的最大预紧力为：

$$F' \leq \frac{\pi d_1^2 [\sigma]}{4 \times 1.3} = \frac{\pi \times 17.3^2 \times 160}{4 \times 1.3} = 28931 \text{ N}$$

根据不滑移公式 $F' \mu z m = K_f R$ 有：

$$R = \frac{F' \mu z m}{K_f} = \frac{28931 \times 0.2 \times 2 \times 2}{1.2} = 19287 \text{ N}$$

- Solution: From the tensile strength condition $\sigma_{ca} = \frac{4 \times 1.3 F'}{\pi d_1^2} \leq [\sigma]$ of an ordinary bolt, it can be obtained that the maximum allowable preload force of the bolt is:

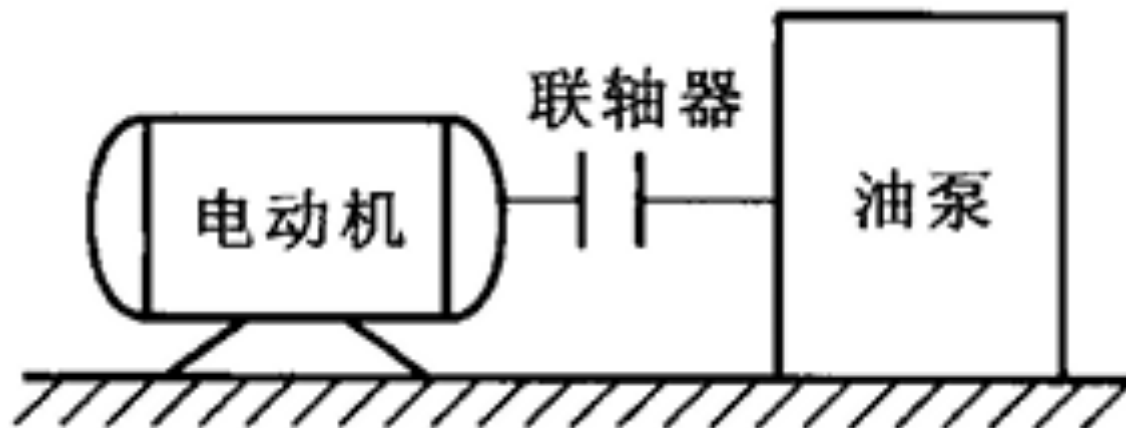
$$F' \leq \frac{\pi d_1^2 [\sigma]}{4 \times 1.3} = \frac{\pi \times 17.3^2 \times 160}{4 \times 1.3} = 28931 \text{ N}$$

- According to the no-slip equation $F' \mu z m = K_f R$ there are

$$R = \frac{F' \mu z m}{K_f} = \frac{28931 \times 0.2 \times 2 \times 2}{1.2} = 19287 \text{ N}$$

HW 05.2

- 如图所示，在电动机与增压油泵间用联轴器相连。已知电动机功率 $P = 7.5 \text{ kW}$ ，转速 $n = 960 \text{ r/min}$ ，电动机伸出轴端的直径 $d_1 = 38 \text{ mm}$ ，油泵轴的直径 $d_2 = 42 \text{ mm}$ ，取 $K_A = 1.7$ 。试确定选用的联轴器是否满足需求：某型弹性圈柱销联轴器的技术参数：许用扭矩： $250 \text{ N}\cdot\text{m}$ ，许用转速： $n_{\max} = 3300 \text{ r/min}$ （联轴器材料为铁）， $n_{\max} = 3800 \text{ r/min}$ （联轴器材料为钢），轴孔直径： $d_{\min} = 32 \text{ mm}$ ， $d_{\max} = 42 \text{ mm}$
- As shown in the figure, a coupling is used to connect the motor and the boost oil pump. It is known that the motor power $P = 7.5 \text{ kW}$, the speed $n = 960 \text{ r/min}$, the diameter of the motor shaft end $d_1 = 38 \text{ mm}$, and the diameter of the oil pump shaft $d_2 = 42 \text{ mm}$. Try to determine whether the selected coupling meets the requirements: Technical parameters of a certain type of elastic ring pin coupling: allowable torque: $250 \text{ N}\cdot\text{m}$, allowable speed: $n_{\max} = 3300 \text{ r/min}$ (coupling material is iron) $n_{\max} = 3800 \text{ r/min}$ (coupling material is steel), shaft hole diameter: $d_{\min} = 32 \text{ mm}$, $d_{\max} = 42 \text{ mm}$



HW 05.2

- 转速和联轴器直径满足要求，计算转矩 $T_{ca} = K_A T$ 。 $K_A = 1.7$ 。

$$\text{名义转矩 } T = 9\,550 \frac{P}{n} = 9\,550 \times \frac{7.5}{960} \text{ N}\cdot\text{m} = 74.6 \text{ N}\cdot\text{m}$$

$$\text{所以 } T_{ca} = 1.7 \times 74.6 \text{ N}\cdot\text{m} = 126.8 \text{ N}\cdot\text{m} < 250 \text{ N}\cdot\text{m}$$

因此该型联轴器符合需求

- The speed and coupling diameter meet the requirements, and the calculated torque $T_{ca} = K_A T$. $K_A = 1.7$.

$$\text{Nominal torque } T = 9\,550 \frac{P}{n} = 9\,550 \times 7.5 / 960 \text{ N}\cdot\text{m} = 74.6 \text{ N}\cdot\text{m}$$

$$\text{So } T_{ca} = 1.7 \times 74.6 \text{ N}\cdot\text{m} = 126.8 \text{ N}\cdot\text{m} < 250 \text{ N}\cdot\text{m}$$

Therefore, this type of coupling meets the requirements

HW 05.3

- 两个半联轴器用四个普通螺栓连接,传递转矩 $T = 1000 \text{ N.m}$, 螺栓分布圆直径为 $D = 300\text{mm}$, 接合面摩擦系数 $\mu = 0.15$, 求螺纹小径 (装配时控制预紧力, 选螺栓材料为 Q235)。
- Two half-coupling with four ordinary bolt connection, transfer torque $T = 1000\text{N.m}$, bolt distribution circle diameter $D = 300\text{mm}$, joint surface friction coefficient $\mu = 0.15$, find the thread diameter (assembly control preload, selected bolt material for Q235).

HW 05.3

- 此连接的工作载荷是转矩 T ，但每个螺栓受横向工作载荷，

- 即： $zR' \times \frac{D}{2} = T$ 螺栓数 $z=4$

螺栓均匀分布，每个螺栓所受的横向工作载荷相等，即： $R' = \frac{2T}{zD}$

每个螺栓所需的预紧力：由摩擦力公式： $m\mu F' \geq K_f R$

$$F' \geq \frac{K_f R'}{\mu m} = \frac{2K_f T}{\mu m z D} = \frac{2 \times 1.2 \times 1000 \times 10^3}{0.15 \times 1 \times 4 \times 300} = 13333.5 N$$

按接合面数 $m=1$ ；可靠性系数 $K_f=1.1 \sim 1.3$ ，取 $K_f=1.2$

按单个螺栓进行强度计算，即

$$d_1 \geq \sqrt{\frac{4 \times 1.3 F'}{\pi [\sigma]}} \text{ mm} \text{ 由表10-6, } [\sigma] = \frac{\sigma_s}{S} \text{ 确定许用应力 } [\sigma]。$$

选择选螺栓材料为Q235，由表10-5选性能级别为 4.6 级， $\sigma_s=240\text{MPa}$ 。

因受控制预紧力，由表10-6， $S=1.2 \sim 1.5$

取 $S=1.4$ ，则 $[\sigma] = \frac{\sigma_s}{S} = \frac{240}{1.4} = 171.4\text{MPa}$

$$d_1 \geq \sqrt{\frac{4 \times 1.3 F'}{\pi [\sigma]}} = \sqrt{\frac{4 \times 1.3 \times 13333.5}{\pi \times 171.4}} = 11.35\text{mm}$$

HW 05.3

- The working load of this connection is the torque T. However, each bolt is subjected to a lateral working load:

$$zR' \times \frac{D}{2} = T \quad z=4$$

The bolts are uniformly distributed and each bolt is subjected to an equal lateral working load:

$$R' = \frac{2T}{zD}$$

Preload force required for each bolt : $m\mu F' \geq K_f R$

$$F' \geq \frac{K_f R'}{\mu m} = \frac{2K_f T}{\mu m z D} = \frac{2 \times 1.2 \times 1000 \times 10^3}{0.15 \times 1 \times 4 \times 300} = 13333.5N$$

$$m=1; K_f=1.1 \sim 1.3, \text{ 取 } K_f=1.2$$

According to table10-6, $[\sigma] = \frac{\sigma_s}{S}$ 。 Selection of selected bolt material Q235, according to table10-5, $\sigma_s=240\text{MPa}$ 。 Due to controlled preload, $S=1.2 \sim 1.5$

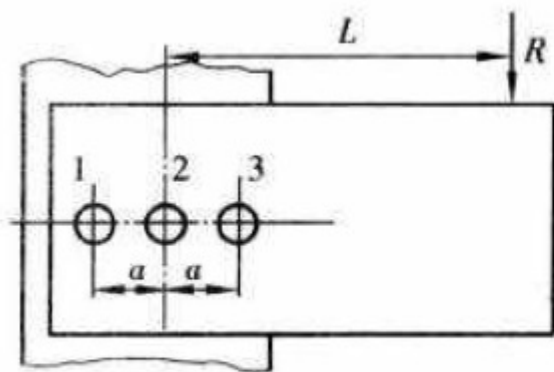
$$\text{set } S=1.4, \text{ then } [\sigma] = \frac{\sigma_s}{S} = \frac{240}{1.4} = 171.4\text{MPa}$$

Strength calculations are made on the basis of individual bolts

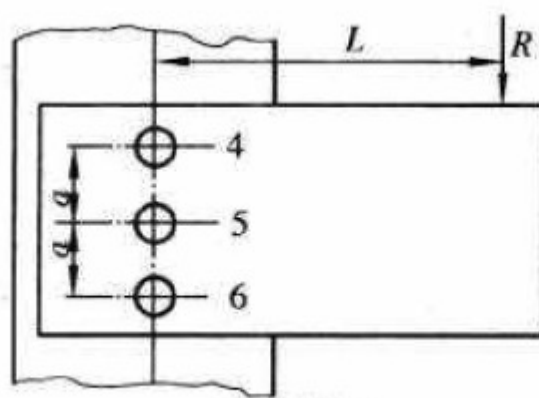
$$d_1 \geq \sqrt{\frac{4 \times 1.3 F'}{\pi [\sigma]}} = \sqrt{\frac{4 \times 1.3 \times 13333.5}{\pi \times 171.4}} = 11.35\text{mm}$$

HW 05.4

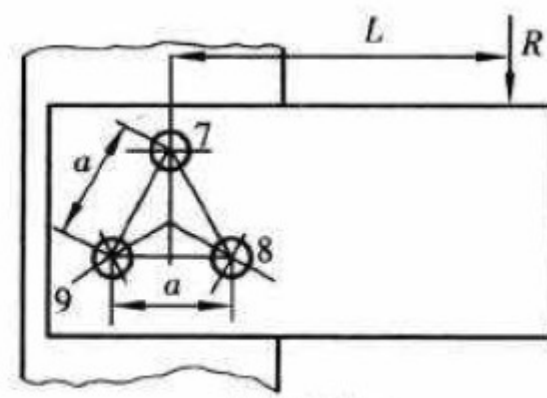
- 图所示为一螺栓组连接的3种方案，其外载荷 R , 尺寸 a 、 L 均相同， $a=60\text{mm}$, $L=300\text{mm}$ 。试分别计算各方案中受力最大螺栓所受横向载荷 F_s ，并分析比较哪个方案好。
- The figure shows three schemes for a bolt group connection, whose external load R , dimensions a and L are the same, $a=60\text{ mm}$, $L=300\text{ mm}$. Try to calculate the transverse load F_s of bolts with maximum force respectively in each scheme, and analyze and compare which scheme is better?



(a) 方案一



(b) 方案二



(c) 方案三

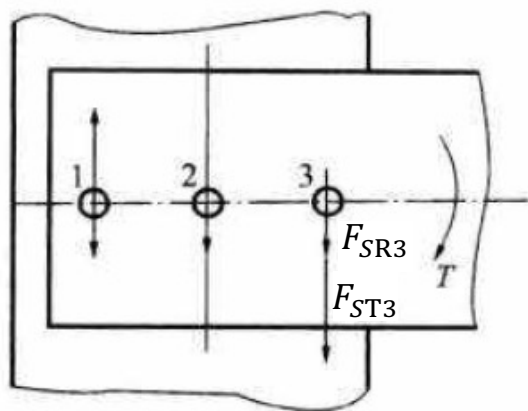
HW 05.4

- 把外载荷 R 向螺栓组连接的接合面形心简化,则该螺栓组连接受有横向载荷 R 和旋转力矩 $T=RL$ 作用。在三个方案中,横向载荷 R 使螺栓组中的每个螺栓受到的横向载荷 F_{sw} 相等,都等于 $R/3$,且具有相同的方向;但由于螺栓布置方式不同,旋转力矩 T 使三个方案中的每个螺栓受到的横向载荷 F_{ST} 却不同。因此,在三个方案中受力最大螺栓所受的横向载荷是不同的。
- If the external load R is simplified to the centroid of the joint surface connected by the bolt group, then the bolt group is subjected to transverse load R and rotating torque $T=RL$. In the three schemes, the transverse load R makes the transverse load F_{sw} of each bolt in the bolt group equal, equal to $R/3$, and has the same direction; However, due to the different bolt arrangement, the rotating torque T causes the transverse load F_{ST} to be different for each bolt in the three schemes. Therefore, in the three schemes, the maximum stress bolt is subjected to different lateral loads.

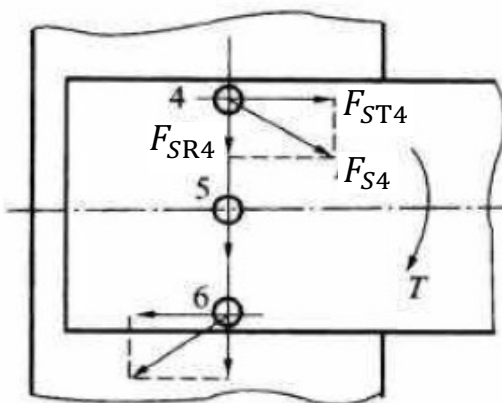
HW 05.4

- (1) 方案一由图 (a) 可知, 螺栓3受力最大, 所受横向载荷为
- (1) According to Figure (a), bolt 3 has the largest force and the transverse load is

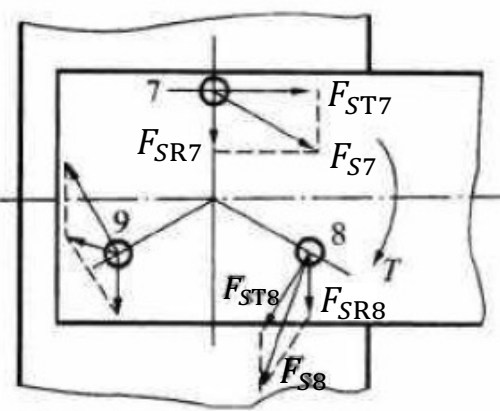
$$F_{S3} = F_{SR3} + F_{ST3} = \frac{R}{3} + \frac{RL}{2a} = \left(\frac{1}{3} + \frac{L}{2a} \right) R = 2.833R$$



(a) 方案一



(b) 方案二

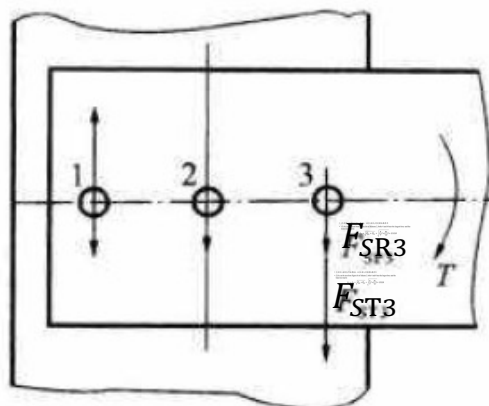


(c) 方案三

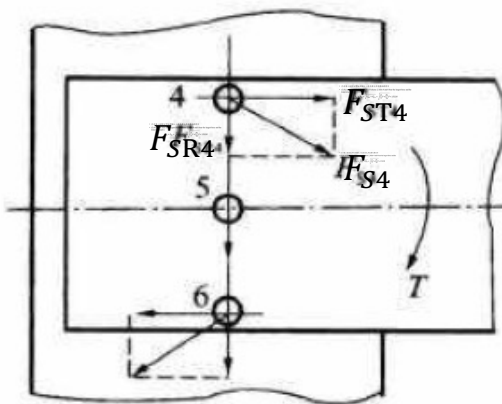
HW 05.4

- (2) 方案二由图 (b) 可知, 螺栓4、6受力最大, 所受横向载荷为
- (2) As can be seen from Figure (b) of Scheme 2, bolts 4 and 6 bear the largest force and the transverse load is

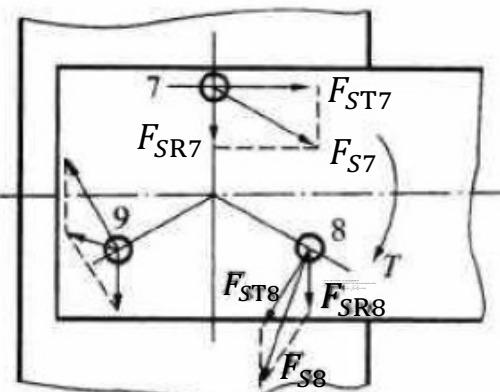
$$F_{S4} = F_{S6} = \sqrt{F_{SR4}^2 + F_{ST4}^2} = \sqrt{\left(\frac{R}{3}\right)^2 + \left(\frac{RL}{2a}\right)^2} = 2.522R$$



(a) 方案一



(b) 方案二

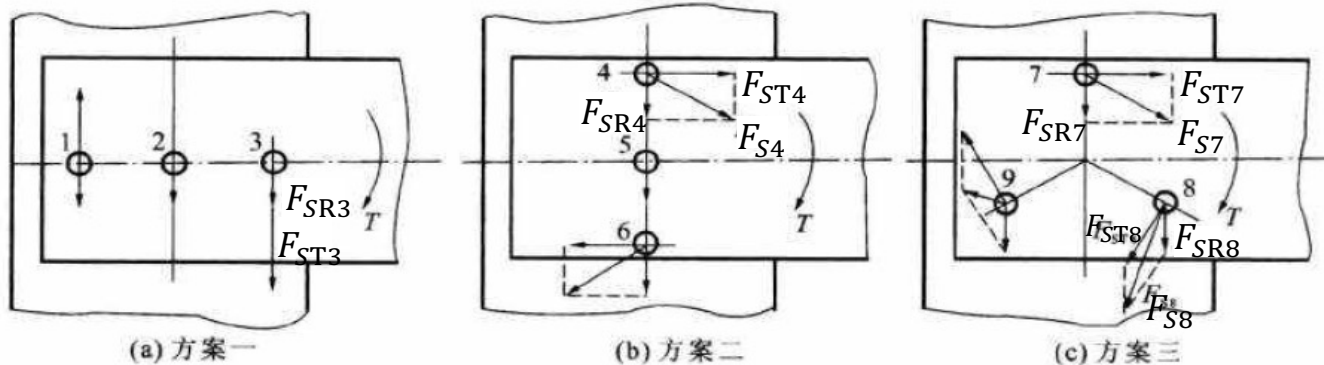


(c) 方案三

HW 05.4

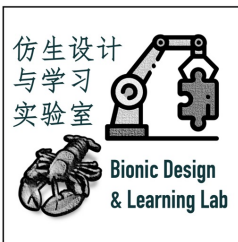
- (3) 方案三由图 (c) 可知, 螺栓8受力最大, 所受横向载荷为
- (3) According to Figure (c), bolt 8 has the largest force and the transverse load is

$$F_{S8} = \sqrt{F_{SR8}^2 + F_{ST8}^2 + 2F_{SR8}F_{ST8} \cos \alpha} = \sqrt{\left(\frac{R}{3}\right)^2 + \left(\frac{RL}{3a}\right)^2 + 2 * \frac{R}{3} * \frac{RL}{3a} * \cos 30^\circ} = 1.962R$$



比较三个方案中受力最大螺栓的受力, 显然方案三中受力最大的螺栓受力最小, 而且从受力分析图中可以看出, 方案三中的三个螺栓受力较均衡, 因此方案三较好。

Comparing the stress of the bolts with the largest force in the three schemes, it is obvious that the bolt with the largest force in the third scheme has the smallest force, and it can be seen from the stress analysis diagram that the three bolts in the third scheme have more balanced forces, so the third scheme is better.



机械设计

Mechanical Design

Thank you~

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